SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Computer Vision with OpenCV

COURSE CODE: DSA0202

COURSE OUTCOME COVERED
CO2: *Apply image processing and feature extraction techniques, including edge detection, motion estimation, and optical flow, for analysing images.. (BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 3 Total Marks:30 Annexure A

Scenario-Based

Code of Conduct:

I K. Pujitha (192524149) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in

disciplinary action. ***Signature of the student: K. Pujitha***

RUBRICS FOR CALCULATION

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Criteria	**Weight age (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Marks**
Understa nding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	

Identifi ca tion of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Applic ati on of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	
Decisi on Makin g	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	
Clarity of Respon se	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

Annexure A

Scenario-Based

1. Crack Detection on Concrete Surfaces Using Drone Images

Scenario: A civil inspection team is developing a computer vision system to detect thin cracks on concrete bridges and building surfaces. Images are captured using a drone under outdoor conditions. Due to changing sunlight, shadows, and camera vibration, the raw images contain uneven illumination, low contrast, and Gaussian noise.

Questions:

A. Propose a suitable image preprocessing pipeline to prepare the drone images for crack detection. Your answer should include methods for illumination correction, noise reduction, contrast enhancement, and image normalization. Justify the need for each step.

B. Select an appropriate edge detection technique for identifying thin surface cracks. Compare it with at least one other edge detection method and explain why your selected technique is more suitable for detecting fine cracks.

C. After edge detection, some crack segments appear broken or disconnected. Suggest a suitable post-processing technique to connect the crack regions and improve the final detection output.

2. Human Motion Tracking in a Security Camera Video

Scenario: A security camera is installed in a hallway to monitor people moving in different directions. The system must estimate the motion of each person across video frames for tracking and activity analysis. The hallway contains plain walls, smooth floors, and occasional shadows.

Questions:

A. Explain how the Lucas-Kanade optical flow method can be used to estimate the motion of people between consecutive video frames.

B. Discuss how the aperture effect can create errors in motion estimation, especially on uniform regions such as plain walls, floors, and clothing surfaces.

C. If a person suddenly changes speed or direction, describe how the optical flow vectors will be affected. Suggest a method to improve tracking accuracy in such situations.

3. Vehicle Tracking at a Traffic Junction Using Video Frames

Scenario: A smart traffic monitoring system uses CCTV footage to track vehicles at a busy road junction. Vehicles move at different speeds, some stop suddenly at signals, and others turn left or right. The video also contains shadows, reflections, and partial occlusion when vehicles overlap.

Questions:

A. Explain how optical flow can be used to estimate the direction and speed of moving vehicles in the video sequence.

B. Identify two challenges that may affect accurate motion estimation in this scenario, such as occlusion, shadows, sudden braking, or camera vibration. Explain how each challenge affects the optical flow result.

C. Suggest a suitable improvement method to make vehicle tracking more reliable when vehicles overlap or change direction suddenly.

1) Crack Detection on Concrete Surfaces Using Drone Images

(A) Suitable Image Preprocessing Pipeline for Drone Images

Answer

Drone images captured from concrete surfaces usually contain uneven illumination, shadows, low contrast, and Gaussian noise. Preprocessing is required to improve image quality before crack detection.

The preprocessing pipeline is:

Input Image → Illumination Correction → Noise Reduction → Contrast Enhancement → Normalization → Crack Detection

1. Illumination Correction

Outdoor images may contain brightness variations due to sunlight and shadows. Techniques such as homomorphic filtering or background correction can be used to reduce uneven illumination.

Purpose:

- Removes shadow effects.
- Provides uniform brightness.
- Makes cracks more visible.

2. Noise Reduction

Since the image contains Gaussian noise, a Gaussian filter is applied.

Purpose:

- Removes random noise caused by camera vibration.
- Prevents false crack detection.
- Maintains important crack details.

3. Contrast Enhancement

Methods like CLAHE (Contrast Limited Adaptive Histogram Equalization**)** are used to improve contrast.

Purpose:

- Enhances the difference between cracks and concrete.
- Highlights weak and thin cracks.

4. Image Normalization

Pixel intensity values are converted into a standard range (0–255).

Purpose:

- Reduces variations between images.
- Improves consistency for further processing.

Thus, preprocessing improves crack visibility and increases detection accuracy.

(B) Suitable Edge Detection Technique for Thin Crack Detection

Answer

The most suitable edge detection method for detecting thin cracks is the Canny Edge Detector.

Canny edge detection involves:

1. Gaussian smoothing for noise removal.
2. Gradient calculation to find intensity changes.
3. Non-maximum suppression to produce thin edges.
4. Thresholding to identify strong edges.

Comparison with Sobel Edge Detector

Canny	Sobel
Produces thin and accurate edges	Produces thicker edges
Handles noise effectively	More sensitive to noise
Detects weak cracks better	May miss fine cracks
Better edge localization	Moderate accuracy

Canny is more suitable because concrete cracks are thin and irregular, requiring accurate edge localization and noise reduction.

(C) Post-processing Technique to Connect Broken Crack Segments

Answer

After edge detection, some crack regions may appear disconnected due to noise and low contrast. A suitable post-processing method is Morphological Closing.

Morphological Closing

Closing consists of:

Dilation followed by Erosion

Working

- Dilation connects nearby broken crack pixels.
- Erosion restores the original shape.

Advantages

- Connects discontinuous crack regions.
- Fills small gaps in cracks.
- Produces continuous crack boundaries.

Other techniques such as connected component analysis can also be used to remove unwanted regions.

2)Human Motion Tracking in a Security Camera Video

Given Scenario

- A security camera monitors people moving in a hallway.
- The system needs to estimate human motion between consecutive video frames.
- The environment contains:
  - Plain walls

    - Smooth floors
    - Shadows
- The objective is to track human movement accurately using optical flow.

(A) Explain how the Lucas-Kanade optical flow method can be used to estimate the motion of people between consecutive video frames.

Answer

Lucas-Kanade Optical Flow is a motion estimation technique used to calculate the movement of objects between two consecutive video frames.

It is based on the assumption that the intensity of a moving object remains constant between frames.

The optical flow equation is:

Ixu+Iyv+It=0I_xu + I_yv + I_t = 0Ixu+Iyv+It=0

where,

- IxI_xIx = intensity change in x-direction
- IyI_yIy = intensity change in y-direction
- ItI_tIt = intensity change over time
- u,vu,vu,v = motion velocity components

Working Steps

1. **Feature Point Detection**
    - Important points such as corners and edges on a person's body are identified.
    - These points are easier to track between frames.
2. **Motion Estimation**
    - Lucas-Kanade method analyzes a small neighborhood around each feature point.
    - It calculates the displacement of these points from one frame to the next.
3. **Optical Flow Vector Generation**
    - The movement is represented using arrows called optical flow vectors.
    - The direction of the arrow represents motion direction.
    - The length represents motion speed.

Application in Human Tracking

For example:

- If a person walks forward, feature points on the body move in the same direction.
- The calculated motion vectors help the system estimate the person's position in each frame.

Thus, Lucas-Kanade optical flow helps track human movement and analyze activities in surveillance videos.

(B) Discuss how the aperture effect can create errors in motion estimation, especially on uniform regions such as plain walls, floors, and clothing surfaces.

Answer

The aperture effect is a limitation in optical flow estimation where motion cannot be accurately determined from regions with insufficient visual information.

It occurs when an object is viewed through a small window or region.

Effect on Uniform Regions

In a security camera scene:

- Plain walls
- Smooth floors
- Single-colored clothing surfaces

contain very few intensity changes.

Because there are no strong edges or textures, the optical flow algorithm cannot determine the exact direction of movement.

Example

If a person wearing a plain shirt moves:

- The algorithm may detect only brightness changes.
- It cannot identify whether the movement is horizontal, vertical, or diagonal.

Problems Caused

- Incorrect motion vectors.
- Missing feature points.
- Tracking errors.
- Inaccurate speed estimation.

Solution

To reduce aperture effect:

- Use corner features instead of flat regions.
- Combine optical flow with object detection methods.
- Use feature detectors such as Harris Corner Detector or Shi-Tomasi Corner Detector.

Therefore, uniform regions create ambiguity in motion estimation because they lack sufficient image information.

(C) If a person suddenly changes speed or direction, describe how the optical flow vectors will be affected. Suggest a method to improve tracking accuracy in such situations.

Answer

When a person suddenly changes speed or direction, the optical flow vectors will also change.

Effect on Optical Flow Vectors

1. **Change in Speed**

- If the person moves faster, the length of optical flow vectors increases.
- If the person slows down, the vector length decreases.

2. **Change in Direction**

- The direction of the optical flow vectors changes according to the new movement direction.
- Previous motion estimates may become incorrect.

Problems

- Tracking may become unstable.
- Feature points may be lost.
- Incorrect position estimation may occur.

Method to Improve Tracking Accuracy

A suitable method is to combine Lucas-Kanade optical flow with a Kalman Filter.

Kalman Filter

- Predicts the future position of the person based on previous motion.
- Corrects errors using new observations.
- Provides smoother tracking even during sudden movements.

Other methods include:

- Multi-object tracking algorithms.
- Re-detecting lost features.
- Combining optical flow with deep learning-based object detectors.

3) Vehicle Tracking at a Traffic Junction Using Video Frames (10 Marks)

Given Scenario

- A smart traffic monitoring system uses CCTV video to track vehicles at a busy road junction.
- Vehicles have different speeds and movements.
- Some vehicles stop suddenly at signals, while others turn left or right.
- The video contains:
    - Shadows
    - Reflections
    - Partial occlusion
- The objective is to estimate vehicle motion accurately and improve tracking reliability.

(A) Explain how optical flow can be used to estimate the direction and speed of moving vehicles in the video sequence.

Answer

Optical flow is a computer vision technique used to estimate the motion of objects between consecutive video frames.

It calculates the movement of pixels caused by moving vehicles.

The basic assumption of optical flow is that the intensity of a moving object remains approximately constant between two consecutive frames.

The optical flow equation is:

Ixu+Iyv+It=0I_xu + I_yv + I_t = 0Ixu+Iyv+It=0

where:

- IxI_xIx = change in intensity along x-direction
- IyI_yIy = change in intensity along y-direction
- ItI_tIt = change in intensity with time
- u,vu,vu,v = velocity components of the moving object

Working in Vehicle Tracking

1. Feature Detection

Important points such as:

- Vehicle corners
- Edges
- Number plates
- Vehicle boundaries

are detected in the current frame.

2. Motion Estimation

The optical flow algorithm compares the position of these features in consecutive frames.

The displacement of each feature point is calculated.

3. Motion Vector Generation

The movement is represented using motion vectors.

- The direction of the vector indicates vehicle movement direction.
- The length of the vector indicates vehicle speed.

For example:

- Long vector → Fast-moving vehicle.
- Short vector → Slow-moving vehicle.
- No vector → Stationary vehicle.

Thus, optical flow helps estimate the speed and direction of vehicles and supports traffic monitoring applications.

(B) Identify two challenges that may affect accurate motion estimation in this scenario. Explain how each challenge affects the optical flow result.

Answer

Two major challenges affecting optical flow-based vehicle tracking are:

1. Occlusion

Explanation

Occlusion occurs when one vehicle partially or completely blocks another vehicle from the camera view.

Effect on Optical Flow

- Feature points of the hidden vehicle may disappear.
- The algorithm may lose track of the vehicle.
- Incorrect motion vectors may be generated.
- Vehicle identity may be confused when it appears again.

Example

A car behind a bus may not be visible for some frames, causing tracking failure.

2. Shadows and Reflections

Explanation

Vehicles moving under sunlight produce shadows, and shiny vehicle surfaces create reflections.

Effect on Optical Flow

- Shadows and reflections create false intensity changes.
- The algorithm may consider them as moving objects.
- Incorrect optical flow vectors are generated.
- False vehicle detection may occur.

Other Possible Challenges

Sudden Braking

- Vehicle motion changes suddenly.
- Previous motion prediction becomes inaccurate.
- Tracking may become unstable.

Camera Vibration

- The entire scene appears to move.
- The algorithm may confuse camera motion with vehicle motion.

(C) Suggest a suitable improvement method to make vehicle tracking more reliable when vehicles overlap or change direction suddenly.

Answer

A suitable improvement method is to combine optical flow with object detection and Kalman filtering.

1. Kalman Filter

The Kalman filter improves tracking by predicting the future position of vehicles.

Working

- Uses previous vehicle position and speed information.

- Predicts the next position.
- Corrects the prediction using new video observations.

Benefits

- Handles sudden speed changes.
- Reduces tracking errors.
- Maintains vehicle tracking during temporary occlusion.

2. Object Detection Integration

Deep learning-based object detectors such as:

- YOLO
- Faster R-CNN

can be combined with optical flow.

Benefits

- Detects vehicles even after temporary disappearance.
- Maintains vehicle identity.
- Improves multi-vehicle tracking accuracy.

Improved Tracking Pipeline

CCTV Video Frames

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Vehicle Detection

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Optical Flow Motion Estimation

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Kalman Filter Prediction and Correction

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Reliable Vehicle Tracking Output